# Band-1 Primes n²+1: CONDITIONAL GRH / Thin-Set Probe (Collapse 2.26)

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## Band-1 Primes $n^2 + 1$ — CONDITIONAL Probe (Collapse 2.26)

**Honesty stamp.** This note is a **CONDITIONAL** literature + finite-box probe. It does **not** prove Landau's fourth problem (infinitely many primes of the form $n^2 + 1$). It does **not** claim that GRH implies Landau-4. It does **not** claim a Clay-millennial solution or a completed ToE. **GATE-04** is locked out of scope.

> **Label: CONDITIONAL.** ERH / thin-set statements below are labeled as such. **Full Landau-4 remains OPEN** (Band 4).


### Abstract

**Problem (Landau-4).** Are there infinitely many primes $p = n^2 + 1$?

**Literature (unconditional almost-prime).** Iwaniec (*Invent. Math.* **47**, 1978) proved there are infinitely many $n$ such that $n^2 + 1$ has at most two prime factors ($P_2$). That is **not** infinitude of primes.

**Literature (CONDITIONAL thin set).** Under the extended Riemann hypothesis for Hecke $L$-functions, Ankeny and Kubilius obtain infinitely many primes of the form $x^2 + y^2$ with $y = O(\log p)$. That is a **thin-set** near-square result under **ERH** — **not** the $y = 1$ case Landau asked for. No standard theorem asserts GRH ⇒ Landau-4.

**Box.** Enumerated all $n \le N = 10^6$ with $n^2 + 1$ prime: **count** = 54,110 (OEIS A005574 prefix match; HL/Landau–Shanks ratio $\approx 1.09$). Finite count ≠ infinitude.


### Problem (Band-1 collapse item 2.26)

1. State Landau-4 honestly as **OPEN**.
2. Cite Iwaniec $P_2$ as the classical almost-prime weakening (not primes).
3. Cite **CONDITIONAL** ERH thin-set (Ankeny–Kubilius); refuse any GRH ⇒ Landau-4 claim.
4. Count on a feasible box window; write `results.json`.
5. Keep Clay / ToE / GATE-04 off the claim surface.

### Literature ledger

**Unconditional status**

| Statement | Status |
|---|---|
| Infinitely many primes $n^2 + 1$ (Landau-4) | **OPEN** (Band 4) |
| GRH ⇒ Landau-4 | **No standard theorem — not claimed** |
| Infinitely many $n$ with $\Omega(n^2 + 1) \le 2$ | **Proved** — Iwaniec 1978 ($P_2$) |

**CONDITIONAL / thin-set cousins**

| Hypothesis | Conclusion | Authors / ref | Label |
|---|---|---|---|
| **ERH** (Hecke $L$) | Infinitely many primes $x^2+y^2$ with $y=O(\log p)$ | Ankeny; Kubilius | **CONDITIONAL** — not $y=1$ |
| (none) | Primes $x^2+y^2$ with $y=O(p^{0.119})$ | Harman–Lewis | unconditional thin cousin — still not $y=1$ |
| (none) | Infinitely many $n^2+1=P_2$ | Iwaniec, *Invent. Math.* 47 (1978) | almost-prime, not prime |

**Heuristic (not a proof)**

Hardy–Littlewood / Bateman–Horn predict

$$\#\{n \le N:\ n^2+1 \text{ prime}\} \sim C \int_2^N \frac{dt}{\log(t^2+1)}, \qquad C \approx 1.37281346$$

(Landau–Shanks constant). Heuristic agreement is **not** a theorem.

**Claims not elevated**

Preprint / Zenodo "resolutions" of Landau-4 are **CLAIMS only** — not community-accepted; not elevated here.

## Method (box)

1. Odd $n>1$ give even $n^2+1>2$ — discard; keep $n=1$ and even $n$.
2. Deterministic Miller–Rabin bases $[2,3,5,7,11,13,23]$ (sufficient for $n^2+1 \le\sim 10^{12}$).
3. Cap $N=10^6$ for dash wall ($\approx 2.5$ s).
4. Label: `BOX_METHOD_DEMO` — **not** Landau-4.

## Results

**CONDITIONAL status (literature)**

| Quantity | Value | Label |
|---|---|---|
| Full Landau-4 | **OPEN** | honest |
| Iwaniec $P_2$ | proved | almost-prime only |
| Ankeny–Kubilius thin set | under **ERH** | **CONDITIONAL** |
| GRH $\Rightarrow$ Landau-4 | **not claimed** | locked |

**Box window (certified tonight)**

| Quantity | Value |
|---|---|
| $N$ | 1,000,000 |
| Count ($n \le N$, $n^2+1$ prime) | 54,110 |
| HL estimate ($\approx C\,N/(2\log N)$) | $\approx$ 49,684 |
| Count / HL | $\approx 1.089$ |
| Largest $n$ | 999,960 |
| Largest value $n^2+1$ | 999,920,001,601 |

| Quantity | Value |
|---|---|
| Wall (box) | $\approx 2.50$ s |
| OEIS A005574 prefix | **PASS** $(1, 2, 4, 6, 10, 14, 16, 20, 24, 26, 36, 40, \ldots)$ |

**Decade counts ($\#\{n < 10^k : \; n^2 + 1$ prime$\}$)**

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| count | 4 | 19 | 112 | 841 | 6656 | **54110** |

## What this does not claim

- Infinitely many primes of the form $n^2 + 1$ (**Landau-4 OPEN**).
- That GRH or RH implies Landau-4.
- That Iwaniec $P_2$ is a prime-infinitude theorem.
- Any Clay problem or completed ToE.
- Any GATE-04 Object-1 meter (collection **LOCKED**).

## Artifacts

- Lab: `mission-control/experiments/band1/primes_n2_plus_1/` (`count_n2_plus_1.py`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `METHOD.md`)
- Paper/PDF: `papers/2026-09-11_Band1_primes_n2_plus_1_GRH_probe.{md,pdf}`
- Shelf: `/workspace/DOWNLOAD_PDFs/` + Mac `~/Downloads/DOWNLOAD_PDFs_2026-09-11/`

## Return line

$N = 10^6$**, count** $= 54110$**, status = CONDITIONAL (ERH thin-set lit) + finite box; full Landau-4 OPEN.**